Final Project
Design and Performance Characterization of a Detachable Lithium-Ion Battery Pack Using MATLAB and Simulink

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1. Project Overview

The final project focuses on designing and modeling a detachable lithium-ion battery pack for outdoor power equipment, such as lawn mowers, leaf blowers, and snow blowers.

The main idea is to develop a single battery pack that can be used with multiple devices, providing flexibility and convenience for users. The battery should meet the company's performance requirements while maintaining safe operating conditions.

MATLAB and Simulink tools, including the Battery Builder app, are used to design the battery pack, simulate its behavior, and evaluate its performance under different operating conditions.

2. Project Objectives

The main objectives of this project are to:

- Design a battery pack using 18650 lithium-ion cells.
- Determine the appropriate number of cells and their series-parallel configuration to meet the required specifications.
- Incorporate thermal modeling to monitor battery temperature.
- Simulate the battery under ideal operating conditions to verify that it meets the minimum performance requirements.
- Evaluate charging and discharging behavior at different current levels and temperatures.
- Characterize battery performance to provide useful information for a consumer user manual.

3. Project Phases

The project is divided into three main phases:

Part 1: Battery Pack Design

Design the battery pack using the Battery Builder app. Select the appropriate cell configuration to meet the required specifications and incorporate thermal modeling to monitor temperature.

Part 2: Battery Performance Verification

Simulate the designed battery under ideal operating conditions and verify that it satisfies the company's minimum performance requirements.

Part 3: Battery Characterization Under Different Operating Conditions

Investigate the battery's charging and discharging characteristics under different current levels and temperatures. Analyze the simulation results to understand how operating conditions affect battery performance.

4. Expected Outcome

The expected outcome is a battery pack design that meets the specified requirements, together with simulation results that demonstrate its performance under different operating conditions. These results can be used to characterize battery behavior and support the preparation of a user manual.

Part 1: Battery Module Design

1. Design Requirements

The objective is to maximize the capacity of a battery module using Panasonic UR18650ZTA lithium-ion cells while satisfying the following design constraints:

- **Nominal Voltage:** The battery module must have a nominal voltage of approximately **18 V.**
- **Cell Mass:** The total mass of all cells in the module must be **less than 1 kg**.
- **Maximum Capacity:** Maximize the battery module capacity while satisfying both voltage and mass constraints.

2. Design Calculations

The selected cell is the Panasonic UR18650ZTA, with a nominal voltage of **3.7 V**, a nominal capacity of **2.9 Ah**, and a maximum cell mass of **49 g.**

2.1 Series Configuration

The number of cells connected in series is selected to achieve a nominal voltage close to 18 V.

$$\left[N_s = \frac{18}{3.7} \approx 4.86\right]$$

Therefore, five cells are connected in series:

$$[V_{\text{module}} = 5 \times 3.7 = 18.5 \text{ V}]$$

2.2 Parallel Configuration

The total mass of the cells must remain below 1000 g.

$$\left[N_s N_p m_{\text{cell}} < 1000\right]$$

Substituting the selected values:

$$\left[5 \times N_p \times 49 < 1000\right]$$

Therefore, the maximum feasible number of parallel cells per series group is four.

3. Final Design

The selected configuration is **5S4P**, consisting of five series-connected groups, each containing four parallel-connected cells.

- **Total number of cells:** 20
- **Nominal voltage:** 18.5 V

- **Nominal cell capacity:** 2.9 Ah
- **Nominal module capacity:** 11.6 Ah
- **Maximum total cell mass:** 980 g
- **Approximate nominal energy:** 214.6 Wh

The selected configuration satisfies the nominal voltage and cell mass requirements while maximizing the module capacity under the specified constraints.

4. Nominal and Effective Battery Capacity

Although the cell capacity listed in the Parameterization Manager and Battery Builder app is 2.9 Ah, an effective capacity of **2.65 Ah** is used in the battery model to more accurately capture cell dynamics across a range of operating conditions.

For the 5S4P configuration, the effective module capacity is calculated using the four parallel cells:

$$[C_{\text{effective}} = 4 \times 2.65 = 10.6 \text{ Ah}]$$

The nominal capacity remains 11.6 Ah, while the effective capacity used for modeling is 10.6 Ah.

Using the nominal module voltage of 18.5 V, the corresponding approximate energy values are:

$$[E_{\text{nominal}} = 18.5 \times 11.6 = 214.6 \text{ Wh}]$$

$$[E_{\text{effective}} = 18.5 \times 10.6 = 196.1 \text{ Wh}]$$

The effective capacity can be verified by opening the module block parameter dialog and checking the **Battery Capacity** value.

5. Battery Builder Implementation

The battery module is configured using the following settings:

- **Cell Model:** Battery Equivalent Circuit
- **Cell Parameterization:** Panasonic UR18650ZTA
- **Thermal Model:** LumpedThermalMass
- **Parallel Assembly:** Four parallel cells with StackingAxis = Y
- **Module Configuration:** Five parallel assemblies connected in series
- **Model Resolution:** Lumped
- **Ambient Thermal Path:** CellBasedThermalResistance

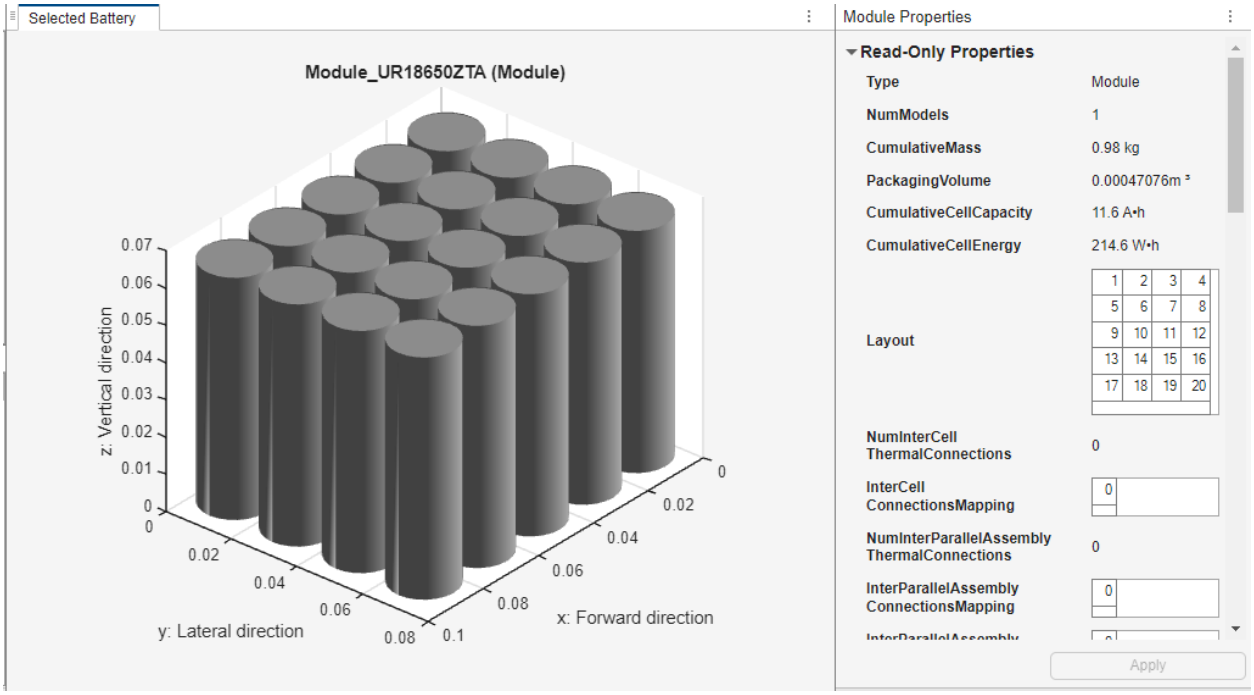


Figure 1: *Battery Builder model of the 5S4P module (20 Panasonic UR18650ZTA cells), with read-only properties showing a cumulative mass of 0.98 kg, 11.6 Ah capacity, and 214.6 Wh energy.*

A library is then created from the battery module itself, not a Module Assembly. The library is used to construct a constant-current discharge circuit for simulation and performance evaluation.

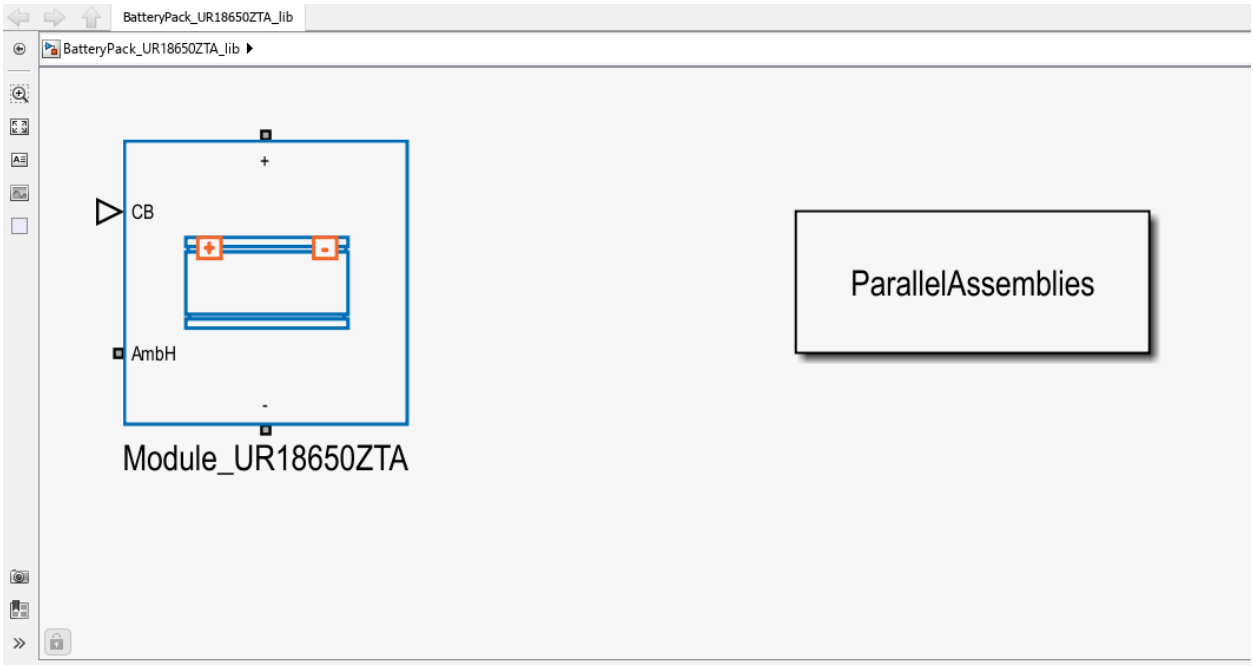


Figure 2: *Simulink library generated from the battery module (Module_UR18650ZTA).*

6. Constant-Current Discharge Circuit

After creating the battery module and generating its library, a constant-current discharge circuit was constructed in MATLAB/Simulink to evaluate the battery's discharge behavior.

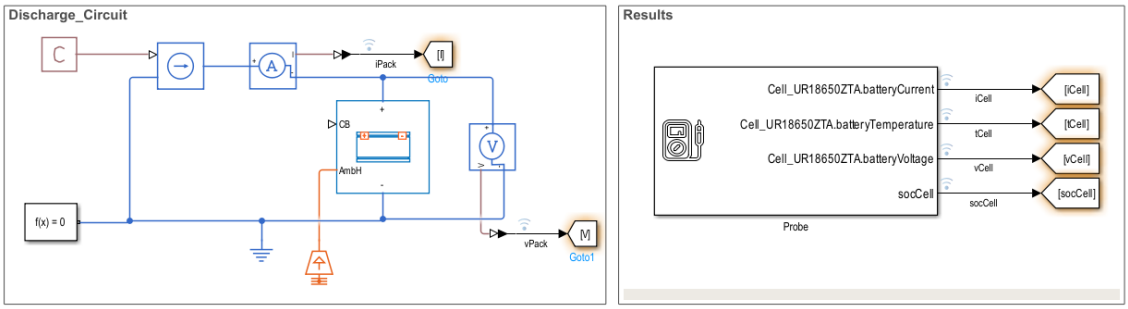


Figure 3: *Constant-current discharge circuit and results (Probe) subsystem in MATLAB/Simulink.*

6.1 Simulation Setup

The battery module has a 5S4P configuration, consisting of five parallel assemblies connected in series, with four cells connected in parallel in each assembly.

The module parameters are:

- **Nominal Voltage:** 18.5 V
- **Effective Capacity:** 10.6 Ah
- **Total Number of Cells:** 20
- **Discharge Rate:** 1C
- **Simulation Duration:** 1 hour

6.2 Constant-Current Discharge

A Controlled Current Source block was used to discharge the battery at a constant current corresponding to a 1C discharge rate.

The required discharge current was calculated using the effective module capacity:

$$\left[I_{\text{discharge}} = \frac{C_{\text{effective}}}{t}\right]$$

For a 1C discharge rate over one hour:

$$\left[I_{\text{discharge}} = \frac{10.6\ \text{Ah}}{1\ \text{h}} = 10.6\ \text{A}\right]$$

The current source input was set to **-10.6 A**. The negative sign indicates that the battery is discharging.

6.3 State of Charge Monitoring

A probe was connected to the battery module to monitor the socCell signal during the simulation.

The simulation was configured to run for one hour. The expected result is that the State of Charge (SOC) decreases from 1 (100%) to 0 (0%), representing a full discharge of the battery at a 1C rate.

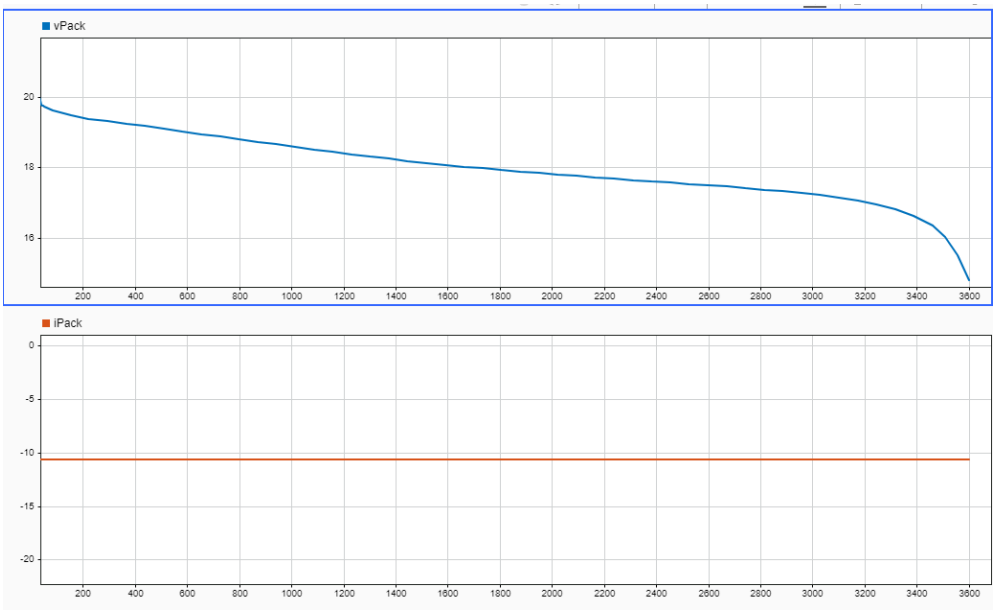


Figure 4: *Pack voltage (vPack) and pack current (iPack) during the 1C constant-current discharge at 10.6 A.*

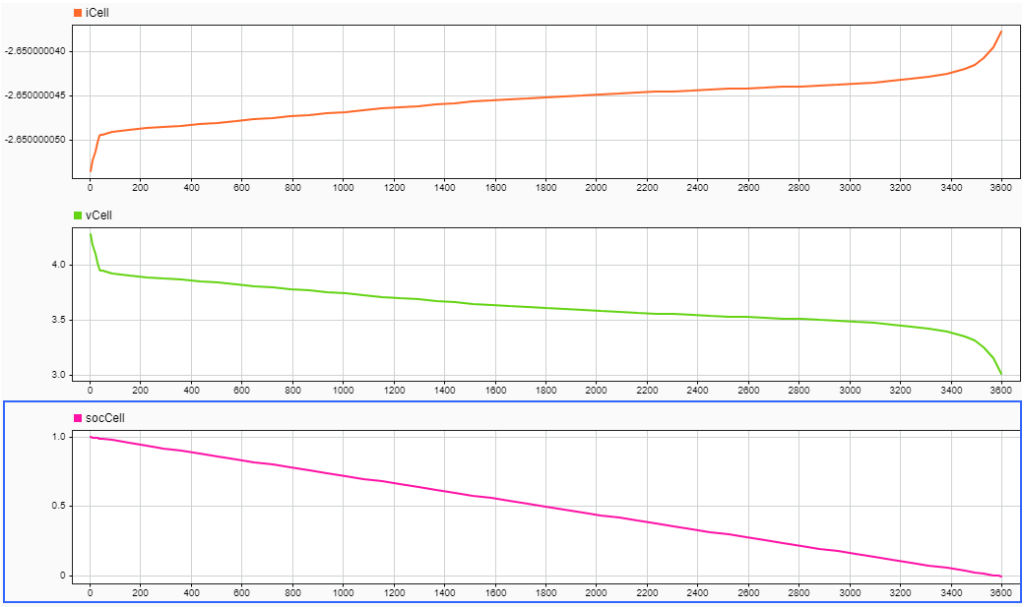


Figure 5: *Cell current (iCell), cell voltage (vCell), and cell state of charge (socCell) during the 1C discharge.*

Part 2: Module Discharge Behavior

1. Objectives and Design Constraints

The objective of this part is to determine the maximum constant discharge power that the battery module can sustain while satisfying the required discharge duration and thermal constraints.

The battery module is initially fully charged and operates at an ambient temperature of 305 K.

The design constraints are:

- **Minimum Discharge Duration:** The module must sustain the selected constant power for at least 30 minutes before being fully discharged.
- **Maximum Battery Temperature:** The module temperature must remain below 330 K throughout the discharge cycle.
- **Maximum Discharge Power:** Determine the highest constant discharge power that satisfies both constraints.

2. Simulation Model

The battery module designed in Part 1 was used to construct a constant-power discharge circuit in MATLAB/Simulink.

A Voltage Sensor was connected across the module to measure its terminal voltage. A Temperature Source was connected to the module's AmbH port and configured to maintain an ambient temperature of 305 K.

A Probe block was used to monitor the socCell and batteryTemperature signals throughout the simulation.

The model was configured to stop when the state of charge moved outside its valid range, as specified in the project instructions.

3. Simulation Procedure

The discharge test was initiated at a constant power of 200 W. The discharge duration and maximum battery temperature were recorded.

The discharge power was then increased in increments of 25 W. Each simulation was evaluated against the minimum discharge duration of 30 minutes and the maximum temperature limit of 330 K.

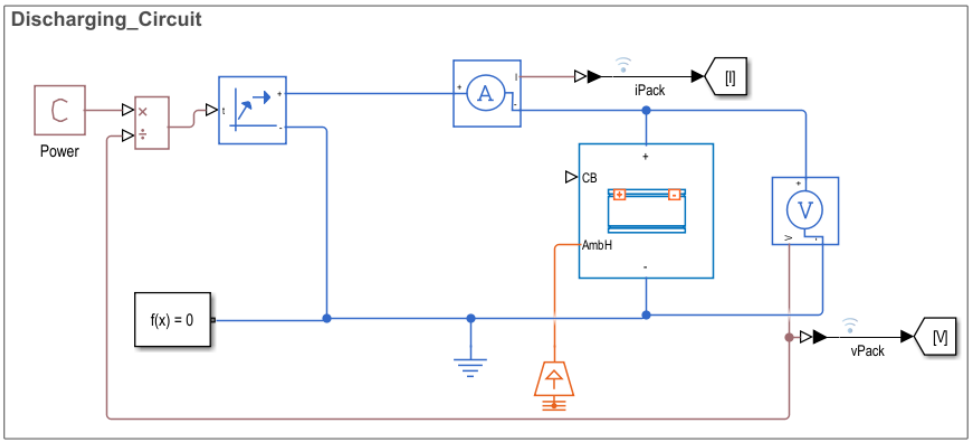


Figure 6: *Constant-power discharge circuit in MATLAB/Simulink.*

4. Results and Analysis

The maximum acceptable discharge power will be identified by comparing the simulation results with both design constraints.

The final results will include the discharge duration, maximum battery temperature, and whether each tested power level satisfies the requirements.

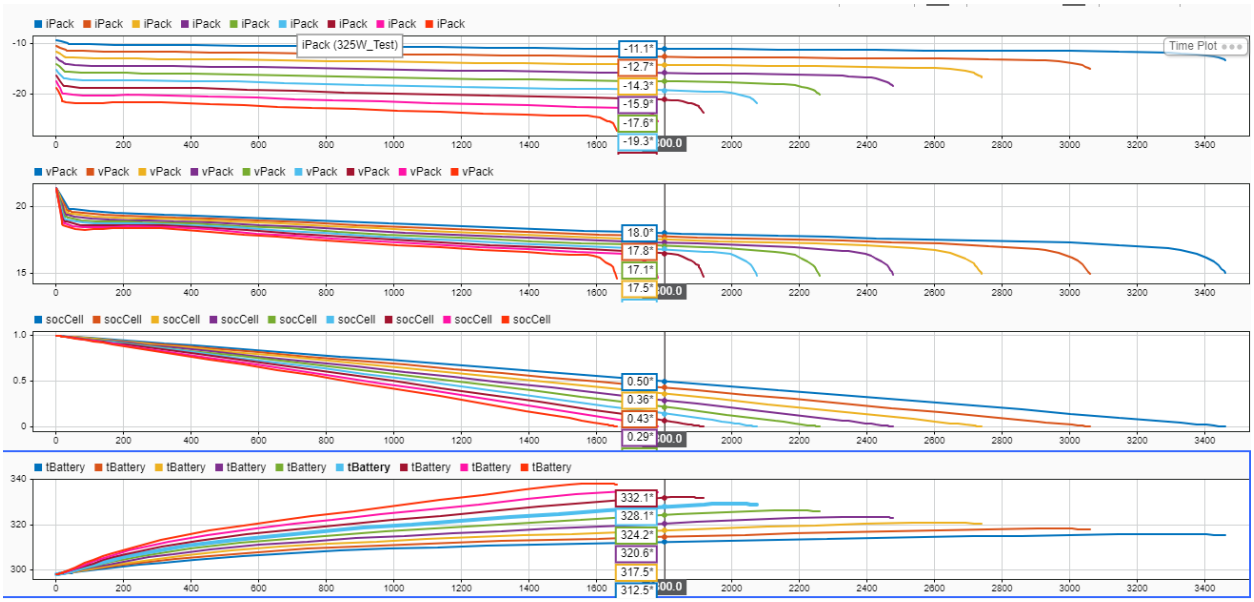


Figure 7: *Constant-power discharge results (pack current, pack voltage, SOC, and battery temperature) for the tested power levels, starting at 200 W in 25 W increments (first set of plots).*

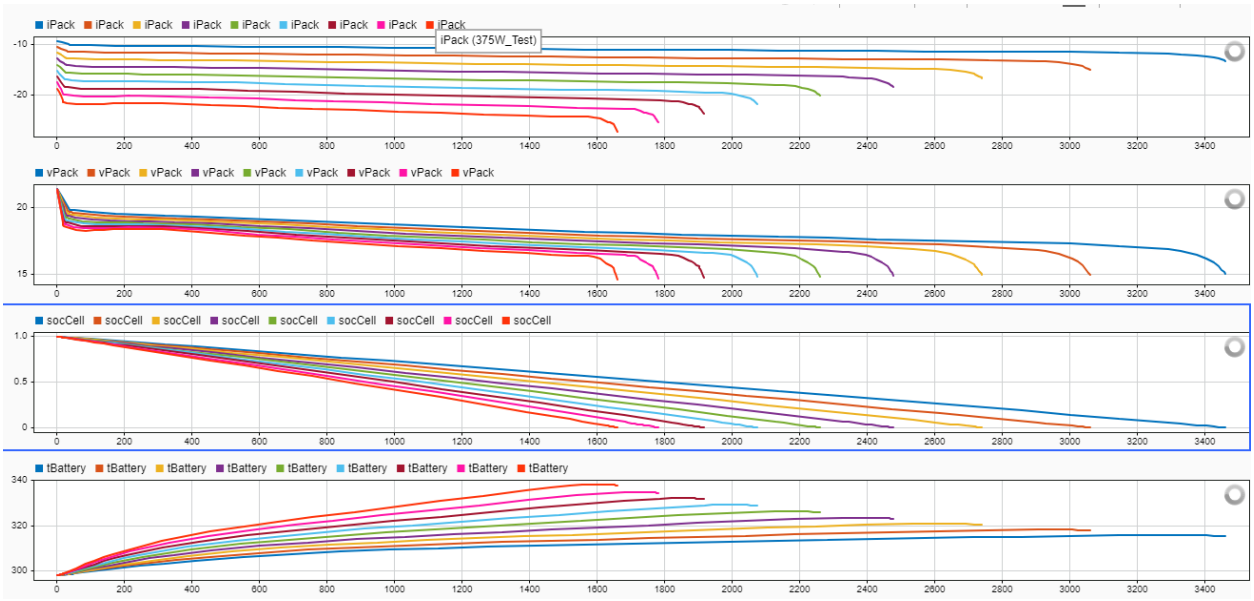


Figure 8: *Constant-power discharge results (pack current, pack voltage, SOC, and battery temperature) for the tested power levels (second set of plots).*

Part 3: CC-CV Battery Charging

1. Objective

The objective of this part is to implement and evaluate a Constant Current–Constant Voltage (CC-CV) charging algorithm for the battery module. The charging process must increase the battery State of Charge (SOC) from 0% to at least 95% within 45 minutes while maintaining the battery temperature below 325 K. The ambient temperature is set to 295 K.

2. Charging Model Configuration

The constant-current discharge circuit developed in the previous part was modified to implement a CC-CV charging algorithm. The Controlled Current Source was replaced with a Charger block from the Cyclers library.

The initial cell SOC was set to 0%, and a Temperature Source was connected to the module's ambient thermal port to maintain the ambient temperature at 295 K. A constant physical signal of 1 was connected to the charger's t input.

The charger was configured using the parameters specified in the project instructions.

Table 1: *Charger block parameters used for the CC-CV charging simulation.*

Parameter	Initial Value
Constant charging current	10.6 A
Voltage threshold	21 V
Constant charging voltage	21 V
Current threshold	0.5 A
Initial SOC	0%
Ambient temperature	295 K

3. CC-CV Charging Algorithm

The charging algorithm operates in two stages:

- **Constant Current (CC) stage:** The charger supplies the specified constant charging current until the battery voltage reaches the voltage threshold of 21 V.
- **Constant Voltage (CV) stage:** The charger maintains the battery voltage at 21 V while the charging current gradually decreases. Charging continues according to the charger settings until the current reaches the specified threshold of 0.5 A.

This method allows the battery to charge rapidly during the initial stage while controlling the voltage as the battery approaches a high SOC.

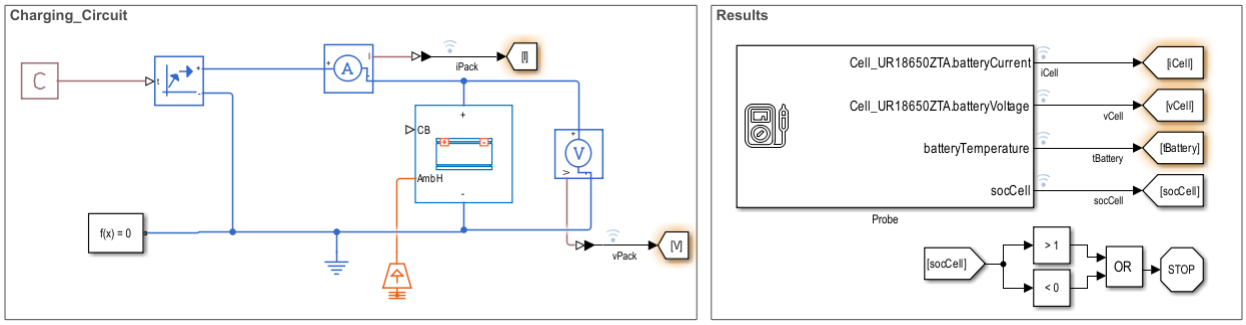


***Figure 9**: CC-CV charge circuit in MATLAB/Simulink.*

4. Simulation Procedure and Results

4.1. Experiment 1: Initial Charging Current of 10.6 A

The first simulation was performed using the initial charging current of 10.6 A, as specified in the project instructions. The simulation duration was set to 2700 seconds (45 minutes).

The SOC and battery temperature were monitored throughout the charging process. The results are summarized below.

***Table 2:** Results of Experiment 1 (constant charging current of 10.6 A).*

Parameter	Result
Constant charging current	10.6 A
SOC at 2700 s (45 min)	75%
Maximum battery temperature	305.8 K
Required SOC	At least 95%
Maximum permitted temperature	325 K

The initial simulation was used to evaluate whether the specified charging current satisfied both the charging-time requirement and the thermal constraint. The results indicated whether further adjustment of the charging current was necessary.

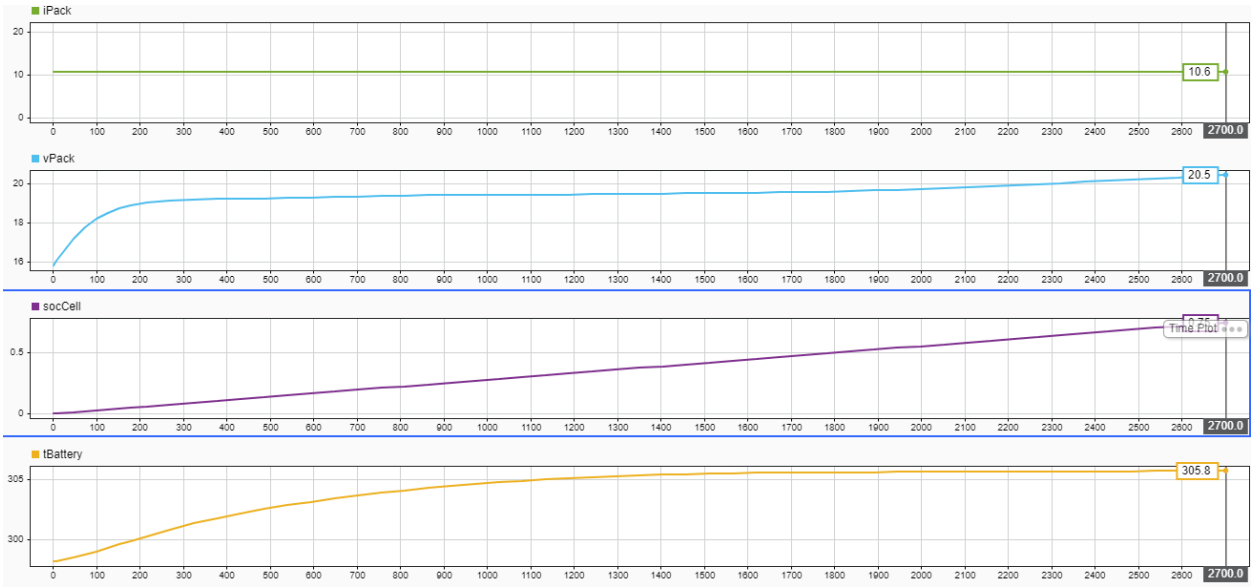


Figure 10: *Experiment 1 (10.6 A): pack current, pack voltage, SOC, and battery temperature over 2700 s of CC-CV charging.*

4.2. Experiment 2: Adjusted Charging Current of 16 A

Following the initial simulation, the constant charging current was increased to 16 A to improve the charging performance. The voltage threshold, constant charging voltage, current threshold, and ambient temperature were kept unchanged.

The simulation was repeated for 2700 seconds, and the SOC and maximum battery temperature were recorded.

Table 3: *Parameters and results of Experiment 2 (constant charging current of 16 A).*

Parameter	Result
Constant charging current	16 A
Voltage threshold	21 V
Constant charging voltage	21 V
Current threshold	0.5 A
Ambient temperature	295 K
Simulation duration	2700 s (45 min)
SOC at 2700 s	95%
Maximum battery temperature	314.8 K
Maximum permitted temperature	325 K

The results show that the battery reached an SOC of 95% within 45 minutes. The maximum temperature was 314.8 K, which is 10.2 K below the permitted limit of 325 K. Therefore, the 16 A charging-current setting satisfied both specified performance requirements under the simulated conditions.

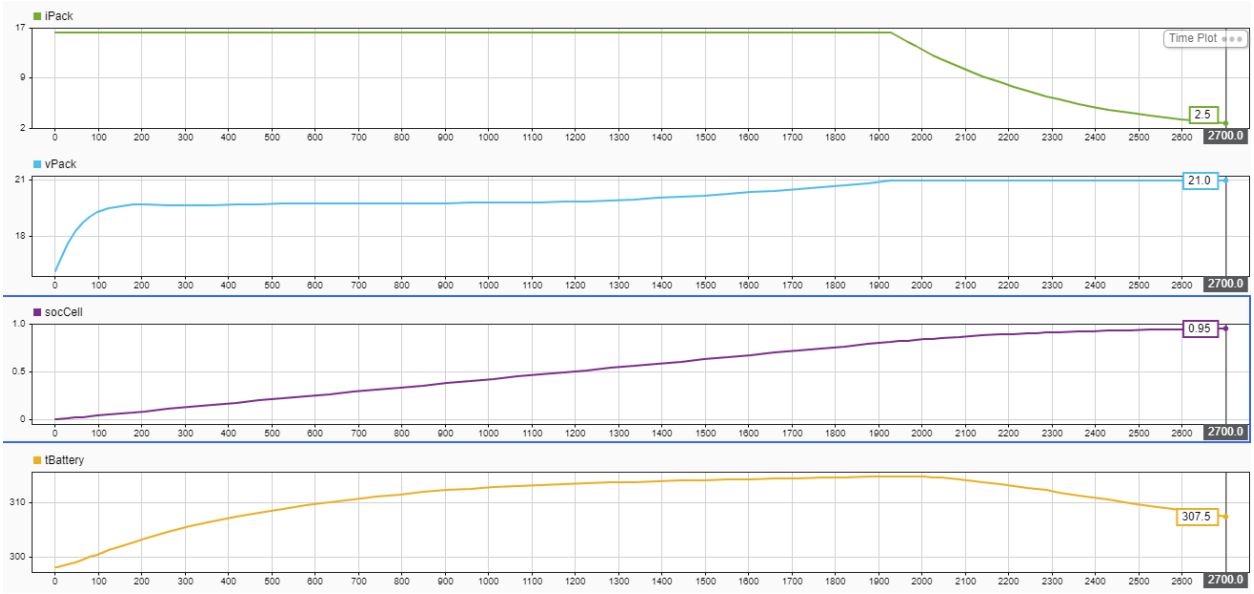


Figure 11: *Experiment 2 (16 A): pack current, pack voltage, SOC, and battery temperature over 2700 s of CC-CV charging.*

5. Comparison and Discussion

The two experiments were used to evaluate the effect of the charging current on battery charging performance. The first experiment followed the specified initial setting of 10.6 A, while the second used an increased current of 16 A.

Increasing the charging current improved the charging performance, allowing the battery to reach the target SOC of 95% within the required 45-minute period. Furthermore, the maximum temperature recorded in the second experiment remained below the specified thermal limit.

The comparison demonstrates the importance of selecting an appropriate charging current to satisfy both charging-time and temperature constraints. However, additional simulations with lower current values would be required to determine whether 16 A is the minimum current that meets both requirements.

6. Conclusion

A CC-CV charging algorithm was implemented using the Charger block in Simulink. The initial simulation used the specified charging current of 10.6 A, with a voltage threshold and constant charging voltage of 21 V and a current threshold of 0.5 A.

The charging current was subsequently increased to 16 A. Under this setting, the battery reached 95% SOC within 2700 seconds, while the maximum battery temperature was 314.8 K. These results satisfy the required charging time and thermal constraints. Therefore, the 16 A setting was identified as a feasible charging-current value for the modeled battery module.

Overall Conclusion

This project covered the design and simulation of a detachable lithium-ion battery pack for outdoor power equipment using MATLAB, Simulink, and the Battery Builder app. The work was carried out in three parts: module design, discharge verification, and CC-CV charging.

- **Part 1 (Module design):** A 5S4P module of 20 Panasonic UR18650ZTA cells was selected. It provides a nominal voltage of 18.5 V, a nominal capacity of 11.6 Ah (10.6 Ah effective capacity used in the model), and about 214.6 Wh of nominal energy, with a total cell mass of 980 g, below the 1 kg limit. A 1C constant-current discharge at 10.6 A took the state of charge from 100% to 0% over one hour, as expected.
- **Part 2 (Discharge behavior):** Constant-power discharge simulations were run from 200 W in 25 W steps at an ambient temperature of 305 K and compared against the 30-minute minimum duration and the 330 K temperature limit. The results show that higher discharge power shortens the discharge time and raises the peak battery temperature, so the maximum usable power is set by whichever constraint is reached first.
- **Part 3 (CC-CV charging):** The initial charging current of 10.6 A reached only 75% SOC after 45 minutes. Increasing the current to 16 A brought the battery to 95% SOC within 2700 s, with a maximum temperature of 314.8 K, which is below the 325 K limit.

Overall, the modeled battery pack meets the specified voltage and mass requirements, and the simulations give useful information on how discharge power and charging current affect run time and temperature. This information can support the consumer user manual. The results come from simulation with a lumped thermal model, so experimental validation would be needed to confirm them. For charging, additional simulations at lower currents would be required to find the minimum current that satisfies both the time and temperature requirements.